Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	12 October 2026
Team ID	SWUID2026403219
Project Name	IMovies – Online Movie, Event & Entertainment Ticket Booking System
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

Functional Requirements – IMovies Ticket Booking System

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	**User Authentication & Authorization**	Register with name, email, mobile and password; verify via email link / 6-digit OTP
		Login with email or mobile (JWT access + refresh token), Remember Me, logout
		Forgot / reset password with time-limited single-use token
		Role-based access: Guest, Customer, Cinema Manager, Organizer, Admin
		Account lockout for 15 minutes after 5 failed logins
FR-2	**User Profile Management**	View and edit profile, change password, upload profile image
		Set preferences: genres, languages, preferred city, notification channels
		Access booking history, favorites and reviews; deactivate account
FR-3	**Location Management**	Detect city by browser geolocation with manual fallback
		Select / change city from popular cities list or search
		Filter movies, cinemas, shows and events by city; list nearby cinemas
FR-4	**Movie Discovery & Details**	Home sections: Recommended, Now Showing, Coming Soon, Popular, Top Rated, Trending
		Movie details page: poster, trailer, cast, director, rating, reviews, cinemas and showtimes

		Global search with auto-complete by movie, actor, director, genre, cinema, event, location
		Filter by language, genre, format, certification; sort by popularity, rating, release date, name
FR-5	**Cinema, Screen & Show Management**	Add, edit, deactivate cinemas and assign Cinema Managers
		Configure screens and seat layouts (Regular, Premium, VIP, Recliner, Couple)
		Create, edit, cancel shows with per-category pricing and overlap check
		View per-show occupancy; booking closes at show start

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-6	**Interactive Seat Selection & Locking**	Visual seat map with category colours and seat states (Available, Selected, Booked, Blocked, Locked)
		Select multiple seats (default max 10); availability refreshed every 5 seconds
		Atomically lock selected seats for 10 minutes with visible countdown timer
		Auto-release locks on expiry, payment failure or cancel
FR-7	**Booking**	Booking flow: movie, cinema, date, showtime, seats, F&B, coupon, order summary, payment, ticket
		Order summary with ticket price, convenience fee, GST, F&B, discount and grand total
		Booking statuses: Pending, Confirmed, Cancelled, Failed, Refunded, Expired, Completed
		My Bookings page (upcoming and past); idempotent booking creation
FR-8	**Payment (Mock Gateway)**	Pay via simulated UPI, credit/debit card, net banking or wallet
		Mock gateway initiate, verify and refund; HMAC-signed idempotent callback
		On success confirm booking, convert seat locks and send notifications; on failure release seats with Retry
		Server-side amount calculation; no card data stored

FR-9	**Food & Beverage**	Cinema-wise menu by category with veg / non-veg flag
		Add, remove, change quantity of items during booking (optional)
		Manager/admin add, edit, disable F&B items
FR-10	**Coupons & Offers**	Apply one coupon per booking with server-side validation
		Percentage or fixed discount with limits, validity, eligibility and applicability
		Offers page; admin create, edit, deactivate coupons and view usage
FR-11	**E-Ticket & QR Code**	Generate signed unique QR code per booking
		Confirmation page; download PDF, print, share, add to calendar (.ics)
		Staff scan / enter booking ID to verify; ticket valid for one use
FR-12	**Cancellation & Refund**	Cancel confirmed booking up to 30 minutes before show; view refund breakup first
		Policy-based refund slabs; convenience fee non-refundable; seats released for resale
		Refund to original method via mock gateway; booking becomes Refunded
		Auto 100% refund when a show is cancelled; admin force-cancel and manual refund
FR-13	**Events Module**	Event categories (concerts, comedy, sports, etc.) with listing and details

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
		Ticket types (General, Silver, Gold, Platinum, VIP) with atomic capacity check
		Organizer creates, publishes, cancels events and views sales and revenue
FR-14	**Ratings, Reviews & Favorites**	Rate and review movies, events, cinemas (verified bookers, one review per item)
		Edit / delete own review; admin moderation and profanity filter
		Add / remove favorites and view Favorites page

FR-15	**Notifications & Recommendations**	In-app notification centre and email for confirmation, payment, cancellation, refund, reminders, new releases
		Asynchronous emails with 3 retries; opt-out for promotional messages
		Personalised recommendations from bookings, genres, languages, popularity and rating
FR-16	**Admin, Manager & Organizer Dashboards**	Admin KPIs: users, movies, cinemas, bookings, revenue, cancellations, active shows
		Manage movies, cinemas, shows, events, banners, coupons, F&B, reviews, users and roles
		View, filter, export bookings and payments; process refunds
		Audit log of all admin actions
		Analytics charts: daily bookings, monthly revenue, popular movies and occupancy

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The UI shall be responsive on mobile, tablet and desktop (320 px and above), follow WCAG 2.1 AA basics, show clear field-level validation, and let a booking be completed in at most 8 steps from the movie page.
NFR-2	**Security**	All traffic over HTTPS/TLS 1.2+; passwords hashed with BCrypt; short-lived signed JWTs; server-side RBAC on every endpoint; protection against OWASP Top 10; signed QR tokens; no card data stored; rate limiting and CORS allow-list.
NFR-3	**Reliability**	Booking and payment operations are ACID transactions with rollback; stuck pending payments reconciled every 5 minutes so a paid booking is never lost; global exception handling returns standard errors without stack traces.

NFR-4 Performance First contentful paint within 3 s on 4G and navigation within 1 s; 95% of read APIs under 500 ms and write APIs under 1 s at 500 concurrent users; seat map loads in 1 s and lock in 500 ms; 200 concurrent attempts on one show with no double booking; paginated and cached lists; compressed, lazy-loaded images.

NFR No.	Non-Functional Requirement	Description

NFR-5	**Availability**	Target 99.5% availability for the deployed demo (excluding planned maintenance); health endpoint (/actuator/health) for monitoring and load balancers.
NFR-6	**Scalability**	Stateless backend for horizontal scaling behind a load balancer; seat locks in a shared store (Redis or DB); design tolerates 100,000 users, 500 cinemas, 10,000 shows/day and 50,000 bookings/day through indexing and caching.
NFR-7	**Maintainability**	Layered architecture (controller, service, repository) with DTOs; at least 70% unit-test coverage on services; OpenAPI/Swagger documentation; Flyway/Liquibase migrations; configurable lock TTL, max seats, fees, tax rate and refund slabs.
NFR-8	**Compatibility & Portability**	Runs on latest two versions of Chrome, Firefox, Edge and Safari; backend and database containerised with Docker Compose.